

Rocío Ávalos Morillas

📄 Biomedical Engineer — Computational Neuroengineering — Clinical Systems

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🌐 LinkedIn

🐙 GitHub

👤 Profile

Biomedical Engineer working at the intersection of AI, neuroimaging, and clinical decision support. Experienced in developing end-to-end pipelines for CT, MRI, and EEG/SEEG data, including deep learning-based segmentation, electrode localization, and multimodal integration. Used to close collaboration with clinicians to translate research prototypes into robust, clinically meaningful software for epilepsy diagnostics.

📁 Experience

Biomedical Engineer – Multimodal Clinical Neuroimaging Systems 09/2025 – Present
BrainFocus, Barcelona (Full-time)

- Contributor to R&D of clinical-grade software for epilepsy, designing end-to-end multimodal systems integrating MRI, CT, and EEG/SEEG data.
- Designed and implemented algorithms for signal processing, electrode localization, and spatial brain mapping across heterogeneous modalities.
- Embedded machine learning models within clinically constrained workflows to improve signal detection robustness and electrode localization accuracy.
- Developed interactive visualization components to support clinician reasoning, presurgical planning, and hypothesis evaluation.
- Translated advanced neuroimaging and signal-processing methods into validated, regulatorily aware, clinician-facing software components.

Machine Learning Research Assistant – SEEG Electrode Segmentation & AI Modeling
02/2025 – 09/2025

Hospital del Mar Research Institute & Center for Brain and Cognition (UPF)

- Conducted applied research on automatic segmentation and localization of SEEG electrodes from post-operative CT scans.
- Developed a 3D Slicer extension for brain segmentation, electrode trajectory estimation, and mask filtering.
- Built ML models to classify true vs. false electrode detections and designed threshold-learning algorithms to generalize across patients.
- Collaborated closely with clinical and neuroscience teams to align the pipeline with surgical planning workflows.

Healthcare Data Analyst Intern 10/2024 – 01/2025
Hospital del Mar, Barcelona

- Managed data inventory and storage for operational efficiency.
- Streamlined data flow between clinicians and software systems.
- Organized medical device inventory and generated analytical reports using hospital information systems.
- Used Jira for project tracking and task management.

SensUs Student Competition – Biomedical Engineer 11/2023 – 08/2024
Eindhoven University of Technology

- Developed a colorimetric microfluidic biosensor for real-time creatinine detection.
- Contributed to sensor prototyping and image-based signal analysis in a multidisciplinary team.

Management Team Member 11/2023 – 06/2025
Bioneer UPC, Barcelona

- Contributed to the technical development and strategic planning of biomedical engineering projects (AI, microfluidics, biosensors).
- Organized events and guided engineering efforts for student competitions.

Education

Master of Science in Advanced Biomedical Technologies 09/2025 – 10/2026

Polytechnic University of Catalonia (UPC)

Focus areas: Computer Vision, Biomedical Data Analytics, Computational Modeling, Bioinformatics

Bachelor of Science in Biomedical Engineering 09/2021 – 06/2025

Polytechnic University of Catalonia (UPC)

Relevant coursework: Medical Imaging, Deep Learning, Signal Processing, Sensors, Medical Devices, Systems Biology

Bachelor's Thesis: Automatic SEEG Electrode Segmentation & Trajectory Reconstruction in 3D Slicer

Grade: 10/10 (Highest Honors) — github.com/rociavl/SEEG_automatic_segmentation

Technical Skills

- ✓ **Programming:** Python, MATLAB, R, SQL
- ✓ **Deep Learning:** PyTorch, TensorFlow, scikit-learn, MONAI
- ✓ **Medical Imaging:** 3D Slicer, SimpleITK, NiBabel, CT & EEG/SEEG processing
- ✓ **Generative AI:** GANs, VAEs, latent space modeling, synthetic data augmentation
- ✓ **Data Science:** Statistical modeling, EDA, predictive modeling, classical ML
- ✓ **HPC & Version Control:** Git, cluster computing fundamentals
- ✓ **Neuroimaging:** SEEG electrode localization, brain mask segmentation, multimodal data fusion

Certifications

- ★ **Neural Networks and Deep Learning** – DeepLearning.AI (2025)
- ★ **Bioinformatics Specialization (with Honors)** – UC San Diego
- ★ **Genome Sequencing (with Honors)** – UC San Diego
- ★ **Advanced Cambridge English Certificate (C1)**

Languages

- 🌐 Spanish, Catalan – Native
- 🌐 English – Advanced (C1, Cambridge Certificate)
- 🌐 French – Intermediate

Selected Projects

SEEG Automatic Segmentation 🧠 — Deep learning pipeline for SEEG electrode localization in CT scans, integrated into clinical workflow.

3D Brain Mask Segmentation 🧠 — UNet model for brain extraction from low-contrast CT.

Synthetic Patient Cohort Generation 🧠 — GAN-based augmentation for rare disease studies.

Chest X-Ray Pathology Classification 🧠 — CNN with transfer learning on CheXpert.

Kidney Disease Prediction & Biomarker Discovery 🧠 — ML pipeline for CKD detection and feature selection.

Research Interests

Generative AI in Healthcare • Synthetic Biomedical Data • Computational Neuroscience • Medical Image Synthesis • Patient Stratification • Clinical Decision Support • Translational Research